Question ID: 16631d34

Question

While researching a topic, a student has taken the following notes:

- The Million Song Dataset (MSD) includes main audio features and descriptive tags for popular songs.
- Audio features include acoustic traits such as loudness and pitch intervals.
- Many algorithms use these audio features to predict a new song's popularity.
- These algorithms may fail to accurately identify main audio features of a song with varying acoustic traits.
- Algorithms based on descriptive tags that describe fixed traits such as genre are more reliable predictors of song popularity.

The student wants to explain a disadvantage of relying on audio features to predict a song's popularity. Which choice most effectively uses relevant information from the notes to accomplish this goal?

Answer

A. Many popularity-predicting algorithms are based on a song's audio features, such as loudness and pitch intervals.

B. Algorithms based on audio features may misidentify the main features of a song with varying acoustic traits, making such algorithms less reliable predictors of popularity than those based on fixed traits.

C. Audio features describe acoustic traits such as pitch intervals, which may vary within a song, whereas descriptive tags describe fixed traits such as genre, which are reliable predictors of popularity.

D. The MSD's descriptive tags are reliable predictors of a song's popularity, as the traits they describe are fixed.